

Regions of abnormally low proton temperature in the solar wind (1965–1991) and their association with ejecta

I. G. Richardson¹ and H. V. Cane²

Laboratory for High Energy Astrophysics, NASA Goddard Space Flight Center, Greenbelt, Maryland

Abstract. Occurrences of solar wind plasma with abnormally low proton temperatures (T_p) have long been associated with the interplanetary manifestations (which we term “ejecta”) of coronal mass ejections (CMEs). We survey the National Space Science Data Center Omni solar wind database for 1965–1991, and data from the Helios 1 and 2 spacecraft for more limited periods, to identify plasma in which T_p is less than the temperature expected (T_{ex}) from the well-established correlation between the solar wind speed and T_p for normal solar wind expansion. The occurrence rate of low-temperature plasma (specifically with $T_p/T_{ex} \leq 0.5$) is shown to be correlated with solar activity levels. Individual low-temperature regions have durations from 1 to ~80 hours with a mean of ~10 hours. Around one third are encounters with the heliospheric plasma sheet (HPS). These events are observed most frequently during the increasing phase of solar activity when the HPS lies closer to the ecliptic. The abnormally low temperatures may be intrinsic to the HPS or may provide support for the proposal that the coronal streamer belt underlying the HPS is a frequent source of ejecta. The remaining events have an occurrence rate which shows a particularly clear correlation with solar activity levels and with the CME rate at the Sun (when CME observations are available), consistent with an association with ejecta. These events also show greater than chance associations with other ejecta signatures. We suggest that the Omni plasma data can provide information on the rate of ejecta passing the Earth, and hence give an indication of the CME rate, for a period commencing before spacecraft coronagraph CME observations became available in the early 1970s. Our findings suggest that T_p depressions may provide a more comprehensive indication of the presence of ejecta than other ejecta signatures, such as bidirectional solar wind electron heat fluxes and energetic ion flows, which alone do not identify all ejecta.

1. Introduction

Plasma in which the proton temperature (T_p) is significantly lower than expected for normal solar wind expansion is one possible manifestation of coronal mass ejections (CMEs) in the solar wind [Gosling, 1990, and references therein]. We call material in the solar wind apparently related to CMEs “ejecta” [e.g., Hundhausen, 1972], reserving the term “CME”, which has recently become applied to this material [e.g., Gosling *et al.*, 1994], strictly for the phenomenon observed near the Sun by coronagraphs. Gosling *et al.* [1973] suggested that abnormally low proton temperatures may indicate expanding regions of plasma which are thermally insulated from the surrounding solar wind, because thermal conduction is less efficient across magnetic field lines. These regions may be closed, plasmoid-like structures disconnected from the Sun or contain magnetic field loops rooted at the Sun into which thermal conduction only occurs from small regions of the corona. The related electron temperature depressions suggested to Montgomery *et al.* [1974] that closed field line structures were present on the

basis of the high electron thermal conductivity. On the other hand, the rapid entry of solar energetic particles into some ejecta [Kahler and Reames, 1991; Farrugia *et al.*, 1993a, b] suggests that field lines in some ejecta are connected to the Sun.

In more recent studies of ejecta material the abnormally low T_p signature has received less attention than others. In particular, bidirectional solar wind electron heat fluxes (BHF) have been advocated as perhaps the best method of identifying ejecta in the solar wind [e.g., Gosling, 1990; Gosling *et al.*, 1992; Phillips *et al.*, 1992; Gosling *et al.*, 1994]. In the work by Richardson and Cane [1993] we examined the occurrence of various ejecta signatures following 40 interplanetary shocks and concluded that no single signature reliably indicates the presence and true duration of ejecta [see Zwickl *et al.* 1983]. However, abnormal T_p depressions were found to be among the more reliable indicators of the presence or absence of ejecta material. To infer when T_p was abnormally depressed, we compared the observed T_p with the temperature expected (T_{ex}) from the well-established correlation between the solar wind speed (V_{sw}) and T_p for normal solar wind expansion [Lopez, 1987, and references therein]. Because of this correlation, abnormally low temperature plasma cannot be identified using a criterion based only on T_p . (Generally, we will use the terms such as “low temperature” or “low T_p ” to mean the value of T_p relative to T_{ex} , not the actual value of T_p .) This was also recognized by Gosling *et al.* [1973] who identified plasma in

¹Also at Department of Astronomy, University of Maryland, College Park.

²Also at Physics Department, University of Tasmania, Hobart, Tasmania, Australia.

VELA spacecraft data from 1965 to 1967 in which T_p was depressed relative to a linear V_{sw} - T_p relationship.

The aim of this paper is to examine further the suggestion that many regions of low- T_p plasma may be associated with ejecta. In particular, we examine whether the occurrence of low- T_p plasma in the inner heliosphere varies with solar activity levels and how low- T_p regions are associated with solar wind structures and with other ejecta signatures. Since extensive T_p measurements have been made from the mid 1960s, they potentially provide information on the presence of ejecta material before the commencement of observations of some other ejecta signatures and of CMEs at the Sun. Also, T_p data are available from numerous spacecraft and may provide a means of investigating ejecta structures using multispacecraft observations. Indeed, in the work by *Cane et al.* [1994] we used near-Earth and Helios 1 and 2 T_p data to study several shock-ejecta ensembles. In the *Cane et al.* [1994] study we also examined the relationship of variations in the count rate of >60 MeV/amu ions to the shock-ejecta structure and concluded that there is a close association between local decreases in the energetic ion density following shocks and ejecta indicated by low- T_p plasma.

The fundamental difference from previous discussions of solar cycle effects in T_p [e.g., *Didato et al.*, 1974; *Feynman*, 1983; *Bruno et al.*, 1994, and references therein] is that we examine the occurrence of solar wind plasma with $T_p \ll T_{ex}$ instead of variations in the mean T_p , which shows little solar cycle dependence. *Freeman and Lopez* [1985] noted that plasma with $T_p < 15,000$ K was more evident at solar maximum. However, they concluded that these cold regions were predominantly associated with slow, normally expanding, solar wind and were not ejecta. We show that plasma with $T_p \ll T_{ex}$ is detected more frequently at higher solar activity levels and tends to occur in events which take several hours to pass by. Around 30% of these events are encounters with the heliospheric plasma sheet (HPS). The occurrence rate of "non-HPS" events is clearly correlated with solar activity levels and the coronagraph CME rate. These events are associated with other ejecta signatures more frequently than would be expected from chance, though since there does not appear to be any other signature which uniquely identifies ejecta, it is not possible to prove conclusively that these low-temperature events are ejecta.

In the next section the identification of abnormally low T_p regions is described. In section 3 we discuss the characteristics of these regions and evidence for a solar cycle dependence in the occurrence rate. The association with ejecta signatures is examined in section 4. In section 5 we show examples of how these regions are related to other ejecta signatures. The observations are summarized in section 6.

2. Data Analysis

The National Space Science Data Center (NSSDC) Omni database [*Couzens and King*, 1986] is a compilation of hourly averaged solar wind observations from near-Earth spacecraft. At the time of writing, T_p data are available from 1965 to early 1991. To calculate T_{ex} ($\times 10^3$ K) from the simultaneously measured solar wind speed, we use a relationship from the *Lopez* [1987] study of a subset of Omni data:

$$T_{ex} = (0.031V_{sw} - 5.1)^2 \quad V_{sw} < 500 \text{ km/s}$$

$$T_{ex} = (0.51V_{sw} - 142) \quad V_{sw} \geq 500 \text{ km/s}.$$

Intervals of abnormally low T_p are then indicated by $T_p/T_{ex} \ll 1$. As discussed by *Lopez* [1987], similar T_p - V_{sw} relationships have been inferred from other data sets [e.g., *Burlaga and Ogilvie*, 1973], suggesting that this relationship is virtually solar cycle invariant. We note though [*Bruno et al.*, 1994] that overlapping observations from more than one spacecraft are not available for part of 1971, so that the Omni plasma parameters before and after this time cannot be cross normalized.

Data from the NSSDC tape of hourly averaged Helios 1 and 2 solar wind data, obtained at 0.3–1.0 AU from the Sun, have been analyzed in a similar way, except that T_{ex} ($\times 10^3$ K) is estimated from a T_p - V_{sw} relationship for Helios plasma data obtained by *Lopez and Freeman* [1986]:

$$T_{ex} = (0.0106V_{sw} - 0.278)^3/R \quad V_{sw} < 500 \text{ km/s}$$

$$T_{ex} = (0.77V_{sw} - 265)/R \quad V_{sw} \geq 500 \text{ km/s}.$$

We divide their relationship by the heliocentric distance R (in astronomical units) to approximate the (slightly V_{sw} -dependent) radial T_p dependence found in their study and by *Totten et al.* [1995]. The data intervals, December 15, 1974, to December 24, 1980 (Helios 1), and January 18, 1976, to March 4, 1980 (Helios 2), encompass the mid 1970s solar minimum to the ~ 1980 solar maximum.

Figure 1 shows T_p versus T_{ex} for the various data sets. The top four panels show two samples of Omni data (1975–1977 around solar minimum and 1979–1982 around solar maximum) and the complete Helios 1 and 2 data sets. In each case the line indicating the least squares fit has a slope of ~ 1 , indicating that the preceding relationships are, on average, valid. However, the shapes of the distributions are determined by data points for which T_p deviates significantly from T_{ex} . The Omni data suggest that this scatter increases at solar maximum. Note that the Helios data extend to higher T_p , because they include observations made at <1 AU. The lower two panels show the ratio T_p/T_{ex} versus heliocentric distance for the Helios 1 and 2 data. The ratio lies around 1 and is virtually independent of heliocentric distance (there is a small residual decrease with heliocentric distance in each case), indicating that the assumption of a $1/R$ dependence in the expression for T_{ex} is a good approximation.

3. Observations of Low-Temperature Plasma Intervals

In this section we discuss the characteristics of regions of abnormally low T_p and examine evidence for a solar cycle dependence in the occurrence rate.

3.1. Examples of Low-Temperature Plasma Intervals

We first show examples of low T_p regions and their relationship to large-scale solar wind structure. Additional events will be illustrated later in the paper [see *Richardson and Cane*, 1993; *Richardson*, 1994; *Cane et al.*, 1994; *Cane and Richardson*, 1995]. Figure 2 shows Omni data for 20 days in October 1974 near solar minimum. The top three panels show the magnetic field intensity, the polar angle, and the azimuthal angle. The next panel shows T_p (solid line) and T_{ex} (dotted line). Shading indicates when T_p is significantly depressed relative to T_{ex} , specifically $T_p/T_{ex} \leq 0.5$. The bottom two panels show the plasma proton density and V_{sw} .

This period is dominated by corotating high-speed solar wind streams in which $T_p \sim T_{ex}$ except at the stream leading

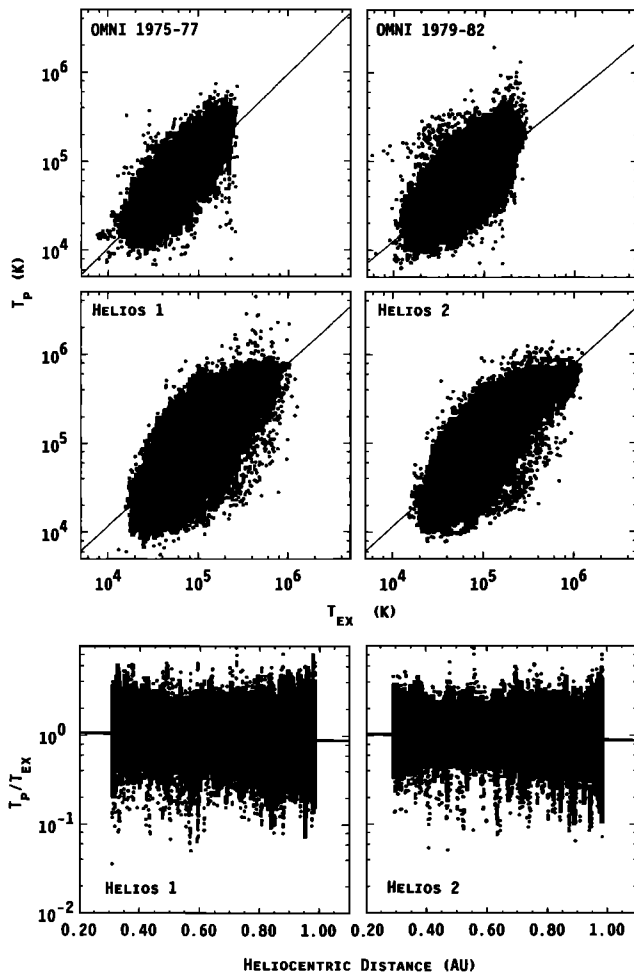


Figure 1. Top four panels show scatter plots of the observed proton temperature (T_p) versus the temperature expected for normal solar wind expansion (T_{ex}) for samples of Omni data near solar minimum and solar maximum and for the Helios 1 and 2 data sets. The bottom two panels show the ratio T_p/T_{ex} for the Helios 1 and 2 data versus heliocentric distance.

edges where $T_p \gg T_{ex}$. This localized heating results from plasma compression in corotating interaction regions (CIRs) [Burlaga *et al.*, 1971], which are indicated by magnetic field and plasma density enhancements at the stream leading edges. There are three regions in which predominantly $T_p/T_{ex} \leq 0.5$. The region on October 12–14 is a magnetic cloud [Klein and Burlaga, 1982]. The region on October 15–16 occurs within a high-speed stream. The associated density enhancement and variations in the magnetic field polar and azimuthal angles, together with the local slowing of the solar wind and depressed T_p , indicate a region of slower, denser, cool plasma embedded in the high-speed stream. The presence of helium abundance enhancements [Borini *et al.*, 1982] and local depressions in the >60 MeV/amu ion count rate [cf. Cane *et al.*, 1994] in both regions, and bidirectional ~ 1 MeV ion flows in the first region [Richardson and Reames, 1993], suggest that these regions are ejecta.

The low T_p region on October 8 has an enhanced plasma density and lies immediately ahead of a separate density enhancement with normal or enhanced T_p associated with a CIR. We interpret this low- T_p region as an encounter with the

heliospheric plasma sheet (HPS). As has been widely discussed [e.g., Borini *et al.*, 1981; Feldman *et al.*, 1981; Gosling *et al.*, 1981; Crooker *et al.*, 1993; Winterhalter *et al.*, 1994], the HPS is characterized by low-speed, dense, cold solar wind with a depressed magnetic field intensity and often one or more magnetic sector boundary crossings, which tends to occur ahead of CIRs but is distinct from plasma in the interaction region. There is another probable HPS encounter on October 22, but in this case, T_p is only slightly depressed relative to T_{ex} . Unfortunately, no magnetic field data are available to help confirm either of these interpretations. However, there are similar features in the Omni data around one 27-day solar rotation period earlier (on September 12–13 and September 24) associated with sector boundary crossings and no field enhancements and indications of corresponding features on earlier and later rotations, which suggest these interpretations are correct. In addition, as discussed below, we find other low- T_p regions for which all data sets are available that are clearly HPS encounters.

3.2. Survey of Low-Temperature Plasma Regions

To illustrate the occurrence of low- T_p plasma over a longer time period, Figure 3 shows Omni, Helios 1, and Helios 2 data during 1974–1979 with T_p/T_{ex} in the range 0.1–0.7. Data points with depressed values of T_p/T_{ex} tend to occur in groups, with combined durations of several hours, rather than randomly. Such groups (events) are more apparent approaching solar maximum in the late 1970s than during the preceding period of lower solar activity (in particular in 1976). We have

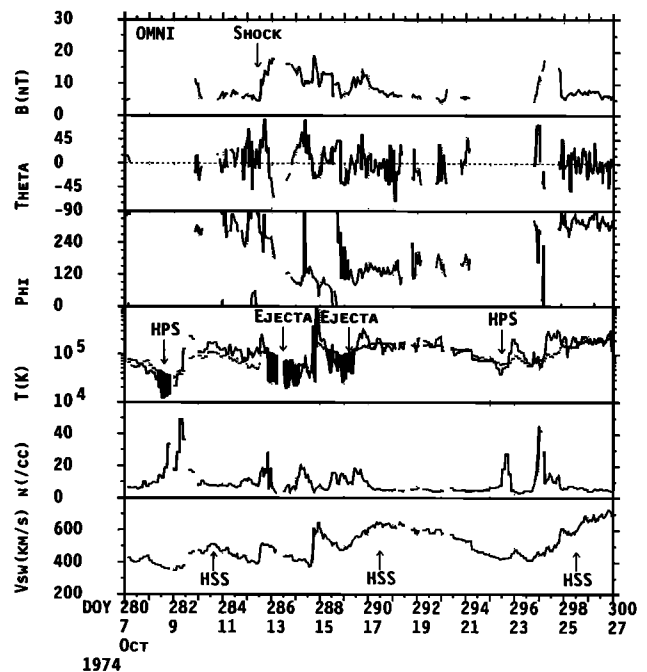


Figure 2. Omni data for October 7–26 1974. The top three panels show the magnetic field intensity, polar angle, and azimuthal angle. The fourth panel compares T_p (solid line) and T_{ex} (dotted line), with shading indicating $T_p \leq 0.5 T_{ex}$. The bottom panels show the plasma density and solar wind speed. This interval is dominated by corotating high-speed streams in which $T_p \sim T_{ex}$. The low- T_p regions on October 12–16 are associated with ejecta, while that on October 8 is an encounter with the heliospheric plasma sheet (HPS).

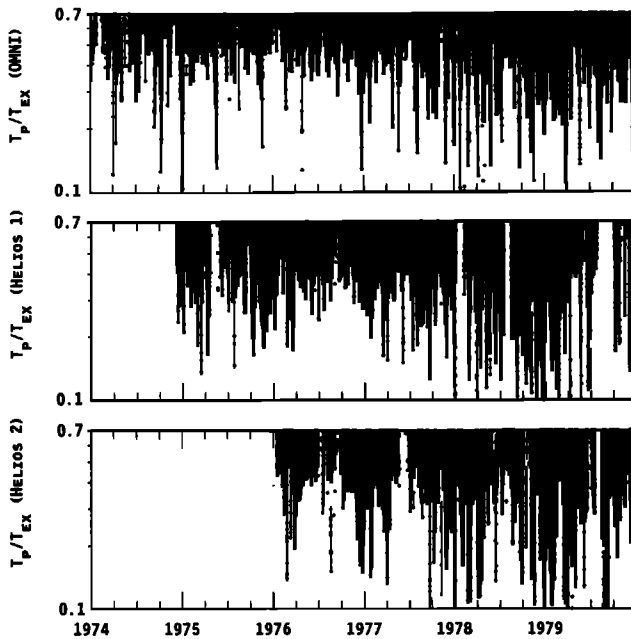


Figure 3. Plots of T_p/T_{ex} during 1974–1979 for Omni, Helios 1, and Helios 2 data. Significant depressions in this ratio with durations of several hours (several data points) are evident throughout this period but are found less frequently during solar minimum conditions in 1975–1977 than during the approach to solar maximum in 1978–1979.

surveyed the Omni and Helios databases to identify intervals when $T_p \ll T_{ex}$. Specifically, we used a criterion of $T_p/T_{ex} \leq 0.5$ based in part on data such as in Figure 3, where this value corresponds approximately to the transition from a general scatter in the data points to discrete low-temperature events. In the Omni data we find 985 events which are evident in one or more data points (i.e., have durations of ~ 1 hour or more). When identifying events, data points with $T_p/T_{ex} \leq 0.5$ separated by less than 3 hours have been grouped together on the assumption that they form part of the same event. In addition, by examining plasma data for each event we have grouped together events which are part of the same temperature depression (i.e., they occur within a longer-duration region where $T_p \ll T_{ex}$ or are separated by brief data gaps).

We find that 273 (28%) of the Omni events are associated with probable HPS encounters. Specifically, they occur in interstream regions of minimum V_{sw} with significantly enhanced plasma density but no increase in magnetic field intensity (frequently there is a field depression and a magnetic sector boundary crossing), often ahead of CIRs. This finding is consistent with the studies cited above, which have shown that the HPS is associated with low- T_p plasma. However, our analysis suggests that T_p in the HPS can be even lower than expected from the low solar wind speed. Considering the Helios data, we identify 799 Helios 1 events of which 278 (35%) are associated with the HPS and 514 Helios 2 events of which 211 (41%) are associated with the HPS. Thus HPS encounters are responsible for a significant minority, around one third, of the low- T_p regions identified.

This result apparently suggests that low- T_p plasma is not a unique signature of ejecta. However, since CMEs nearly always originate from above the coronal streamer belt [Hundhausen, 1993], Crooker *et al.* [1993] have suggested that ejecta may be

observed in the vicinity of the HPS. They suggest that flow from the streamer belt may be quiet, or slowly expanding as small-scale ejecta, or erupting as large-scale ejecta. Thus it is plausible that abnormally low T_p regions associated with some HPS encounters may be signatures of slow ejecta from the streamer belt, rather than intrinsic to the HPS.

We will call the remaining events “non-HPS” events, though this term is not intended to imply that they have no association with the HPS or the underlying streamer belt. For example, a fast ejecta may originate above the streamer belt but would not be classified as an HPS event according to the above criteria. Thus we actually distinguish between low- T_p events associated with encounters with the quiet HPS and those associated with other solar wind conditions.

Figure 4 shows distributions of the Omni and Helios HPS and non-HPS event durations. Only events which do not start or end at a data gap are included. The distributions can be fitted approximately by power laws in duration (t) of the form $N(t) \sim t^{-\mu}$ with μ ranging between 1.0 and 1.2. Considering the non-HPS events, the Omni event durations range from ~ 1 to ~ 80 hours with a mean of 11.5 hours. (The numbers also given in each panel are the mean square deviations from the

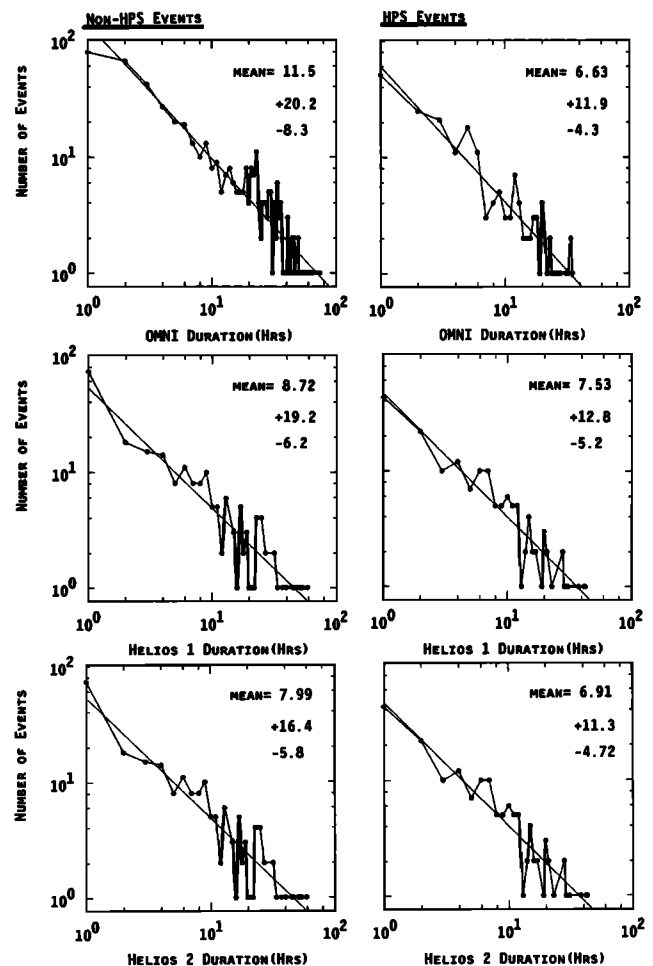


Figure 4. Distributions of the durations of Omni, Helios 1, and Helios 2 events with $T_p/T_{ex} \leq 0.5$. The right-hand column shows distributions for heliospheric plasma sheet events while those for non-HPS events are shown in the left-hand column. The distributions are approximated by power laws in duration (t) of the form $N(t) \sim t^{-1.1}$.

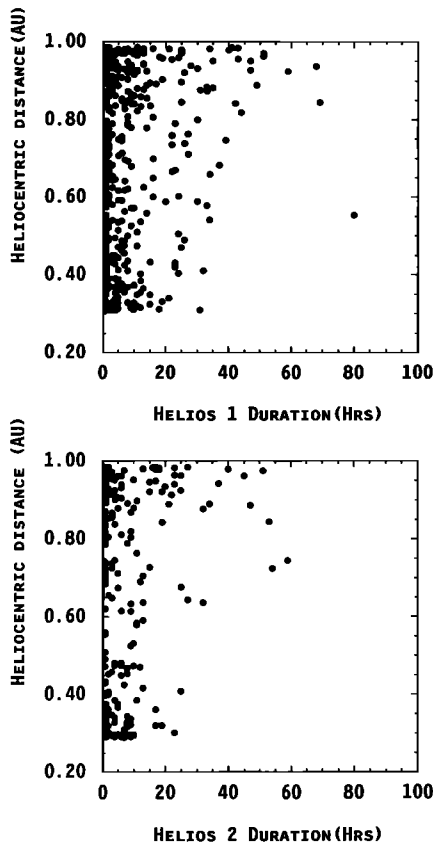


Figure 5. Durations of Helios 1 and 2 non-HPS events plotted versus spacecraft heliocentric distance. There is evidence of a general increase in duration with increasing distance from the Sun.

mean duration for events with durations which are less than or greater than the mean duration and give a measure of the width of the distribution.) Assuming that these regions propagate past the spacecraft at ~ 500 km/s, the event durations imply that the regions extend ~ 0.01 to ~ 0.9 AU (mean ~ 0.1 AU) in the radial direction. The Helios non-HPS events have slightly smaller mean durations (8.7 hours, Helios 1; 8.0 hours, Helios 2), because the durations tend to increase with heliocentric distance (Figure 5). The mean event durations at 0.3–0.4 AU are ~ 5.3 hours (Helios 1) and 4.4 hours (Helios 2), compared to 9.5 hours (Helios 1) and 8.8 hours (Helios 2) at 0.9–1.0 AU. This increase is apparently related to the expansion of these regions as they propagate away from the Sun, as previously inferred from case studies of ejecta at 1 AU [e.g., Farrugia *et al.*, 1993a]. Assuming that these structures move away from the Sun at ~ 400 – 500 km/s, the mean durations imply an increase in the radial extent from ~ 0.05 to ~ 0.09 AU and an expansion rate of ~ 30 – 40 km/s, which is of the order of the Alfvén speed in the solar wind.

The HPS event durations (~ 1 to 50 hours) and means are similar to, though slightly less than, those of non-HPS events. Assuming corotation, the typical duration of ~ 7 hours at all spacecraft is equivalent to a longitudinal extent of $\sim 4^\circ$. On the other hand, the mean radial extent is ~ 0.05 AU, assuming propagation past the spacecraft at ~ 300 km/s.

3.3. Solar Cycle Variations

We now consider the occurrence rate of low- T_p plasma during the solar activity cycle. Although the long-term Omni

database should be ideal to address this question, there are several factors which may influence such an analysis: First, the T_p (plus V_{sw}) data coverage is not complete. The yearly averaged coverage (Figure 6, solid circles) is generally $>40\%$, though occasionally much lower (notably $\sim 9\%$ for 1970). The cadence of the observations also varies. For example, the data after February 1980 are predominantly from IMP 8 and only available at most for the 60% of each 12.5-day period geocentric orbit when the spacecraft was in the solar wind. In contrast, during the preceding period from August 1978, near-continuous data are available from the ISEE 3 spacecraft at the upstream L1 libration point supplemented by data from other spacecraft. Figure 6 also shows the Helios 1 and 2 T_p plus V_{sw} data coverage. We will not attempt to identify and investigate possible systematic effects due to data coverage and cadence and instrumental effects such as changes in sensitivity and calibration, which might be present in these databases. Our intention is merely to see whether the available data allow us to make some general conclusions about possible solar cycle effects in the occurrence of low- T_p plasma.

We have examined yearly averaged occurrence rates (percent) of plasma with $T_p/T_{ex} \leq 1.0$, ≤ 0.5 , ≤ 0.33 , and ≤ 0.2 . Though not shown here, there is little clear solar cycle dependence in the occurrence of plasma with $T_p/T_{ex} \leq 1.0$, where the occurrence rate is $\sim 50\%$. This indicates that there are no strong solar cycle trends in the T_p - V_{sw} relationship, as was also concluded by Lopez [1987]. Figure 7 shows the occurrence rates for $T_p/T_{ex} \leq 0.5$ in the Omni (solid circles), Helios 1 (open circles), and Helios 2 (open squares) data and their relation to solar activity levels as indicated by the yearly averaged monthly mean sunspot number in the top panel (Solar-Geophysical Data, 1965–1991). The error bars assume variances of $\sqrt{N}$, where N is the number of 1-hour averaged data points. No rate is shown for 1991, because the Omni data only cover a period at the beginning of the year when activity was low compared to the general level throughout the year. All the data sets indicate that depressed plasma temperatures are generally found more frequently at solar maximum than at solar minimum. Similar solar cycle variations (not shown here) are found for $T_p/T_{ex} \leq 0.33$ and ≤ 0.2 . However, when a low T_p/T_{ex} threshold is chosen, the rate is dominated by the rela-

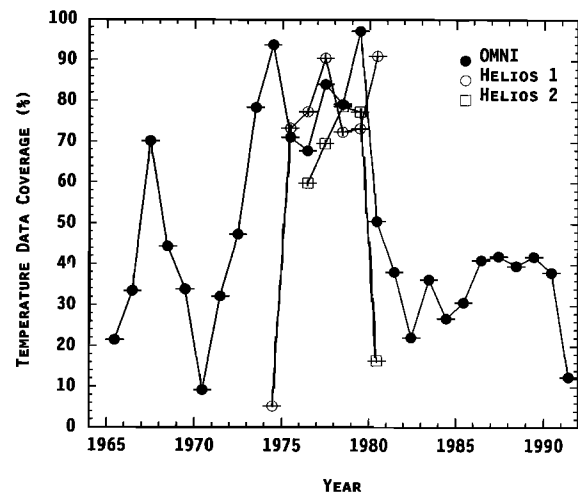


Figure 6. Yearly averaged percentage coverages of the Omni (solid circles), Helios 1 (open circles), and Helios 2 (open squares) proton temperature (plus solar wind speed) data.

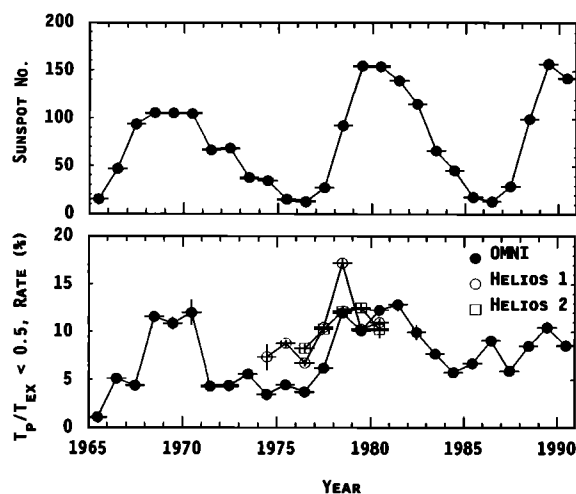


Figure 7. Variation of the yearly averaged sunspot number and occurrence rates of $T_p/T_{ex} \leq 0.5$ plasma in 1965–1990. Solid circles indicate Omni data, open circles Helios 1 data, and open squares Helios 2 data.

tively few events which include very low temperature plasma. On the basis of these considerations we have chosen to use a criterion of $T_p/T_{ex} \leq 0.5$ to identify abnormally low temperature plasma.

In Figure 8 we examine the contributions of HPS and non-HPS events to the $T_p/T_{ex} \leq 0.5$ occurrence rates and their solar cycle dependence. Considering HPS events, the bottom panel of Figure 8 shows the occurrence rate in each data set of low- T_p plasma associated with the HPS. This is obtained by multiplying the occurrence rate of plasma with $T_p/T_{ex} \leq 0.5$ (from Figure 7) by the ratio of the total durations of HPS events and all events in each year. The HPS occurrence rate tends to be anticorrelated with solar activity and be greatest during the increase in solar activity following solar minimum in the mid to late 1970s and mid to late 1980s. *Cliver* [1993, Figure 2] suggests that the tilt angle of the heliospheric current sheet returned toward 0° during these intervals, so that presumably one might have expected the quiet HPS to be encountered more frequently.

The rates for non-HPS events, estimated in a similar way, are shown in the second panel of Figure 8 (note the scale change from the bottom panel). The solar activity dependence is even more marked than in Figure 7 because of the removal of the HPS event contribution. Overall, the Omni and Helios occurrence rates are similar even though the spacecraft heliocentric distances and instrumentation are not identical. Nonetheless, there are some differences. For example, in 1978 the Helios 1 rate was significantly higher than the Helios 2 rate. These differences probably arise because (1) the Helios spacecraft and the Earth may not have encountered exactly the same interplanetary structures, and (2) data gaps in each data set cover different periods, so that even if the data were obtained at similar locations, observations of a particular structure may be missing from a given data set. On the basis of the solar cycle dependencies we suggest that non-HPS events may be closely associated with ejecta, while the prevalence of HPS events when the tilt angle is small suggests that these events most likely correspond to the quiet HPS or to the presence of small-scale, slowly expanding ejections in the HPS, rather than with large-scale ejecta.

It is of interest to estimate the rate (e.g., number of events/year) at which non-HPS events are observed, since this may be related to the rate at which CMEs are ejected from the Sun [Gosling *et al.*, 1992]. Because of data gaps and changing data coverage and cadence, it is difficult to estimate the event rate reliably from the number of individual events detected in a

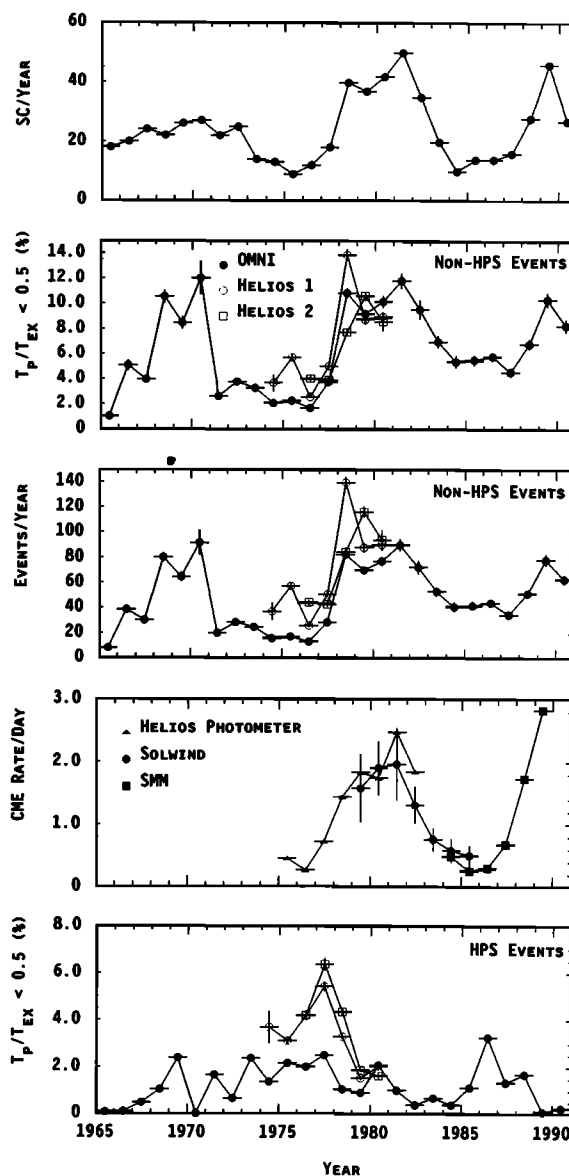


Figure 8. Yearly averaged occurrence rates of plasma with $T_p/T_{ex} \leq 0.5$ associated with HPS (bottom panel) and non-HPS (second panel) events are compared with the number of sc/year in the top panel. The third panel shows the estimated non-HPS event rates (events/year) at each spacecraft based on the occurrence rates in the second panel and the mean event durations in Figure 4. The fourth panel shows the yearly averaged CME rate/day from Webb and Howard [1994] based on Solwind and Solar Maximum Mission coronagraph Helios 1 and 2 90° photometer observations. The non-HPS event rates show good correlations with the rates of sc (except during the late 1960s solar maximum) and CMEs, consistent with an association with ejecta. Low-temperature plasma associated with HPS events occurs predominantly during the increasing phase of solar activity.

given period. Instead, we take the non-HPS occurrence rates in the second panel and multiply them by the number of days in the year in order to estimate the time period in each year when low- T_p plasma would have been observed if data coverage were complete. To estimate the number of events/year, this time is divided by the mean duration of non-HPS events from section 3.2. The estimated event rates are shown in the third panel of Figure 8. For the Omni data the rate ranges from ~ 80 events/year at solar maximum to $\sim 15\text{--}40$ events/year at solar minimum. The Helios rates are in reasonable agreement, though they are generally slightly higher than the Omni rate. However, because of the indirect method used to estimate these rates, these differences in the Omni and Helios rates probably do not indicate any significant radial dependence in the event rate and are consistent with the reasonable proposition that ejecta observed in the inner heliosphere continue to move out to beyond 1 AU.

We now consider how changes in the occurrence of low- T_p plasma compare with variations in the interplanetary shock and coronagraph CME rates. The top panel of Figure 8 shows the number of sc/year in the Solar-Geophysical Data (1965–1991) geomagnetic activity summary reports, which we use as a proxy for the rate of interplanetary shocks. There is a good correlation between variations in the occurrence rates of non-HPS low- T_p plasma and the SC rate, except during the late 1960s solar maximum when the enhancement in the SC rate is relatively less significant than that in the occurrence of low- T_p plasma. This difference suggests that some caution may be necessary in using the sc rate during this period as a proxy for the ejecta-CME rate.

The fourth panel of Figure 8 shows the yearly averaged whole Sun CME rate (per day) inferred by *Webb and Howard [1994]* from Solar Maximum Mission (SMM) and Solwind coronagraph observations and from the Helios 1 and 2 white-light photometers. There is clearly a general correlation between the CME rate and the estimated number of non-HPS low- T_p events. Relating the CME rate to the event rate at the Earth involves considering the CME ejection directions relative to the ecliptic and the CME latitudinal extents, as discussed by *Gosling et al. [1992]*. They used the observed latitudinal widths and locations of SMM coronagraph CMEs to estimate that around 60 ejecta/year would be expected to encounter the Earth at solar maximum and ~ 10 ejecta/year at solar minimum. These rates are slightly lower than the non-HPS event rates shown in the third panel of Figure 8. However, *Webb and Howard [1994]* suggest that SMM only observed $\sim 70\%$ of CMEs, so that if the *Gosling et al. [1992]* rates are underestimated accordingly, these rates should be increased to ~ 85 ejecta/year at solar maximum and ~ 14 events/year at solar minimum in better agreement with our estimates. Thus in spite of the considerable assumptions made when estimating the low T_p/T_{ex} event rates these rates are remarkably consistent with the rate at which CME-associated ejecta might be expected to encounter the Earth. However, the significance of this agreement may be questioned since the coronagraph observations do not show any counterparts to the frequent short-duration (and hence small-scale size) structures which are observed in the solar wind. One surprising point in Figure 8 is a tendency for the non-HPS event rate to be higher in the mid 1980s solar minimum than in the previous solar minima, which does not appear to be evident in the CME rates in the fourth panel. This may indicate a systematic effect in the 1980s (IMP 8) T_p data.

4. Association of Low-Temperature Plasma Events with Ejecta Signatures and Shocks

We now examine in Tables 1 and 2 the association of low- T_p plasma events with various ejecta signatures and with shocks. Table 1 shows the percentages of events which are associated with various possible ejecta signatures [e.g., *Gosling, 1990*], as well as with cosmic ray decreases and interplanetary shocks/sc. We consider three groups of events, namely all events, non-HPS events, and HPS events. For each group, association rates are given for all events and for events with durations ≥ 10 hours. An association means that the low T_p and signature intervals at least overlap, not that they coincide exactly (generally this is not the case, as will be evident from the examples to be discussed below). Errors in the association rates are estimated by considering the effect of a change of one in the number of events associated with a signature and are $\leq 1\%$ unless given in the table. To assess the significance of these association rates, we have compared them with the rates at which chance associations would be expected to occur, calculated as in the appendix. The number of events in each group of events, the chance association rates, and the ratio of the observed/chance association rates are given in Table 3. An observed/chance association ratio $\gg 1$ indicates that the signature is associated with low-temperature plasma more frequently than would be expected from chance, suggesting that the phenomena may indeed be related. A ratio $\ll 1$ indicates that the signature is not associated with the group of low-temperature events under consideration. (For some signatures a comprehensive list of all events has not been compiled, so the chance association rate cannot be estimated.) In Table 1 we indicate those association rates which are not significantly greater than 1.5 times the chance association rate.

Note that any list of ejecta signatures is likely to include periods when no observations of this signature were made. Since these periods are often unspecified, it is not always possible to exclude low- T_p events in these gaps from the analysis, causing the association rate to be underestimated. Another point to remember is that individual ejecta signatures may not be observed in all ejecta [*Zwickl et al., 1983; Richardson and Cane, 1993*] (see also section 5). For example, if a signature is observed generally in $\sim 60\%$ of ejecta, and all low- T_p events are in fact ejecta, then the association rate would still only be $\sim 60\%$. However, it would be incorrect to interpret this rate as implying that only 60% of low- T_p events are ejecta.

Table 2 shows the association rates of events in various studies of possible ejecta signatures, as well as shocks/sc with low- T_p plasma. The first column gives the number of events in each study for which T_p data are available. The second column gives the number of these events associated with $T_p/T_{ex} \leq 0.5$, and the third column gives the association rate, where again it is indicated where this rate is not significantly greater than 1.5 times the chance association rate. (The chance association rates, calculated as in the appendix, and observed/chance association rate ratios are shown in Table 4.) Note that the rates in Table 2 may be less than 100% either if low- T_p regions are not associated with ejecta or if the signature being considered is not uniquely associated with ejecta. The latter case can occur. For example, there is evidence that not all bidirectional ion flows are associated with ejecta [*Richardson, 1994*]. The specific possible ejecta signatures we consider are the following:

Bidirectional energetic (~ 1 MeV) ion flows (BIFs) have

Table 1. Percentages of $T_p/T_{ex} \leq 0.5$ Plasma Events Associated With Ejecta Signatures, Cosmic Ray Decreases, and Shocks

	All events, %		Non-HPS, %		HPS, %	
	All	≥ 10 hours	All	≥ 10 hours	All	≥ 10 hours
Bidirectional ~ 1 MeV Ion flows						
IMP 8/ISEE 3 ^a	36	67	40	67	24	65 $\pm$ 3
Helios 1	27	43	30	45	20	37 $\pm$ 2
Helios 2	29	51	30	53 $\pm$ 2	28	46 $\pm$ 4
ISEE 3 bidirectional solar wind electron heat fluxes						
ISEE 3 electron temperature depression ($T_e \leq 10^5$ K)	27	45	32	50	10 $\pm$ 2 ^b	9 $\pm$ 9 ^b
ISEE 3 local electron temperature depression ^c	45	69	43	68	54 $\pm$ 2	73 $\pm$ 9
Magnetic cloud	71	82	68	81	80 $\pm$ 2	90 $\pm$ 10
He abundance enhancement	7	13	9	15	4	5 $\pm$ 2 ^b
Ion count rate depression > 60 MeV/amu ^c	13	28	19	38	2 ^b	6 $\pm$ 3 ^b
IMP 8	28	55	37	63	8	19 $\pm$ 3
Helios 1	27	47	33	61	15	19 $\pm$ 2
Helios 2	29	54	37	68 $\pm$ 2	19	33 $\pm$ 2
Neutron Monitor Decrease $\geq 4\%$						
	9	18	12	21	2 ^b	2 $\pm$ 2 ^b
Shock/sc within ~ 1.5 days before event						
Omni	18 ^b	36	24	43	4 ^b	4 $\pm$ 2 ^b
Helios 1	17 ^b	32	25	48	3 ^b	2 $\pm$ 2 ^b
Helios 2	16 ^b	32	22	48 $\pm$ 2	8 ^b	7 $\pm$ 2 ^b

^aBIFs with durations > 2 hours. BIFs observed by the Goddard Space Flight Center instruments on IMP 8 and ISEE 3 and by the energetic particle anisotropy spectrometer (EPAS) experiment on ISEE 3 [Marsden *et al.*, 1987].

^bRatio of observed/chance association rates (given in Table 3) is not significantly greater than 1.5.

^cChance association rates not calculated for this signature.

been interpreted as counterstreaming ion beams in regions of closed field lines disconnected from the Sun or looped field lines rooted at the Sun, associated with ejecta in particular. The association rates in Table 1 are based primarily on BIF observations made by the Goddard Space Flight Center (GSFC) instruments on the IMP 8, ISEE 3, and Helios 1 and 2 spacecraft [Richardson and Reames, 1993; Richardson, 1994], since we can better correct for gaps in these data and for periods when the particle fluxes were too low to allow their anisotropy to be determined. Association rates for various sets of BIF events are given in Table 2.

Bidirectional solar wind electron heat fluxes have been interpreted in terms of the presence of similar magnetic field structures [Gosling, 1990]. The BIFs are observed by the Los Alamos National Laboratory (LANL) experiment on ISEE 3, so we only consider the period from August 1978 to December 1982 when the spacecraft was close to the Earth.

We have identified intervals of low electron temperatures for Table 1 by using the LANL ISEE 3 electron temperature data (NSSDC data set 78-079A-01K), which extend from August 1978 to October 1982 and choosing a criterion of $T_e \leq 10^5$ K. We have also examined the electron temperature data for evidence of "local" temperature depressions at the times of low- T_p events. In Table 2 there are T_p data for four of the low- T_e events of Montgomery *et al.* [1974]. In addition, we have surveyed the LANL ISEE 3 electron temperature data for regions with $T_e \leq 6 \times 10^4$ K, the threshold used by Montgomery *et al.* [1974]. The low proton temperature events iden-

tified by Gosling *et al.* [1973] are also included in Table 2 for comparison with our similar analysis.

Magnetic clouds are a subset of ejecta characterized by enhanced magnetic fields with a smooth rotation in direction and low- β plasma (where β is the ratio of the thermal to the magnetic field pressures). Only a minority of ejecta have magnetic cloud signatures (e.g., $\sim 30\%$ of the events discussed by Gosling [1990] and none of the ~ 20 ejecta associated with strong solar events identified by Richardson and Cane [1993]), so the association rates in Table 1 are expected to reflect this.

Solar wind plasma helium abundance enhancements have been reported in some ejecta [e.g., Borrini *et al.*, 1982; Zwickl *et al.*, 1983]. The association rates in Table 1 are based on the helium abundance enhancement surveys of Borrini *et al.* [1982] and Neugebauer and Alexander [1991]. In addition, events from the earlier study of Hirshberg *et al.* [1972] and references therein are considered in Table 2.

We have previously suggested [Cane *et al.*, 1993, 1994] that ejecta may be associated with local depressions in the high-energy particle count rate. In Table 1 we consider the association of low- T_p events with significant depressions in the > 60 MeV/amu particle count rate measured by anticoincidence guards on the GSFC IMP 8 instrument and the University of Kiel experiments on Helios 1 and 2. By significant we mean a depression which exceeds the statistical and point-to-point variations prevailing around the time of the event and which starts and ends close to the boundaries of the low-temperature event. Depressions at higher energies are examined by consid-

Table 2. Association Rates of Ejecta Signatures and Shocks with $T_p/T_{ex} \leq 0.5$ Plasma

Signature	Events With T_p Data	Events With $T_p/T_{ex} \leq 0.5$	Percent Associated With $T_p/T_{ex} \leq 0.5$
Bidirectional ion flows			
Palmer <i>et al.</i> [1978]	5	3	60 ± 20
ISEE 3, 35–1000 keV [Marsden <i>et al.</i> , 1987]	45	42	93 ± 2
IMP 8 (≥ 4 hours) [Richardson and Reames, 1993]	439	207	47
ISEE 3 (≥ 4 hours) [Richardson and Reames, 1993]	196	86	44
Helios 1 (≥ 4 hours) [Richardson, 1994]	94	45	48
Helios 2 (≥ 4 hours) [Richardson, 1994]	31	18	58 ± 3
Bidirectional solar wind electron heat fluxes			
Gosling <i>et al.</i> [1987]	49	35	71 ± 2
Gosling <i>et al.</i> [1990] (large geomagnetic storms)	23	20	87 ± 4
J. T. Gosling (private communication, 1989)	137	86	63
Plasma electron temperature depressions			
Montgomery <i>et al.</i> [1974]	4	3	75 ± 25
ISEE 3 (Aug. 1978 to Oct. 1982) $T_e \leq 6 \times 10^4$ K	47	37	79 ± 2
Plasma proton temperature depressions			
Gosling <i>et al.</i> [1973]	12	11	92 ± 8
Magnetic clouds			
Klein and Burlaga [1982] and Lepping <i>et al.</i> [1990]	57	42	74 ± 2
He enhancement			
Hirshberg <i>et al.</i> [1972] and references therein	15	10	67 ± 7
Borini <i>et al.</i> [1982]	71	39	55
Neugebauer and Alexander [1991]	17	13	76 ± 6
$\geq 4\%$ neutron monitor decreases			
Classes 1 and 3	47	38	81 ± 2
Shocks/sc			
sc (Solar-Geophysical Data Summary Reports, 1965–1991)	362	177	49
Helios 1 shocks	179	104	58^a
Helios 2 shocks	118	73	62^a

^aRatio of observed/chance association rates (given in Table 4) is not significantly greater than 1.5.

ering cosmic ray decreases observed by neutron monitors. The neutron monitor decreases are obtained from the list of Lockwood [1990] and also from the Cane *et al.* [1993] list of $\geq 4\%$ neutron monitor decreases, which we have now extended to cover events during the period for which we have Omni data. (These lists are essentially identical except for a few events.) It should be noted that such decreases are not uniquely associated with ejecta but may also be produced by turbulence following shocks and by corotating high-speed streams. Cane *et al.* [1993] suggested that particle data over a wide range of energies may be used to divide the neutron monitor events into four classes, a classification we have applied to events in the extended list. In Table 2 we only consider “class 1 and 3” neutron monitor depressions, which are expected to be associated with ejecta.

Although not strictly an ejecta signature, we also consider whether low- T_p regions are observed within an interval of 1.5 days following an interplanetary shock. Since some ejecta travel fast enough to drive shocks, a fraction of all low- T_p regions would be expected to be observed following shocks. The presence of a shock at Earth is based primarily on the occurrence of a geomagnetic sudden commencement (Solar-Geophysical Data, 1965–1991), while information on Helios 1 and 2 shocks is obtained from the Helios shock list of R. Schwenn (private communication, 1993).

Overall, the association rates in Table 1 are generally significantly higher than those expected from chance (Table 3), consistent with an association between low- T_p plasma and ejecta signatures. In general, they tend to be highest for non-HPS events with durations greater than 10 hours, though exceptions do occur. There are some interesting differences and similarities in the relationship between HPS and non-HPS

events and the various ejecta signatures: The association rates of HPS events with BHF, high-energy particle depressions, helium abundance enhancements, magnetic clouds, and shocks are similar to or lower than expected from chance, suggesting that these signatures tend not to be associated with the HPS. On the other hand, BIFs and T_e depressions show similar association rates for HPS and non-HPS events. The shock association rates in Table 1 are barely significant except for non-HPS events with durations >10 hours. This result is plausible, since it implies that shocks are generally only generated ahead of larger ejecta. The Helios shock association rates in Table 2 are also not significantly higher than expected from chance (Table 4). However, the rates are consistent with the observation that shock angular widths are greater than those of the corresponding ejecta, so that only around $\sim 50\%$ of shocks are followed by ejecta signatures [e.g., Cane, 1988; Richardson and Cane, 1993]. Otherwise, the association rates for events in the studies of various ejecta signatures in Table 2 are all significantly higher than expected from chance, typically by a factor of 3 or more (Table 4), suggesting a general association between the presence of these ejecta signatures and abnormally low temperature plasma.

5. Examples of the Relationship Between Low-Temperature Plasma and Other Ejecta Signatures

We now illustrate examples of low- T_p events and their association with ejecta signatures. The event on April 25–26, 1979, shown in the Omni data in Figure 9, is an ejecta following a shock at 2358 UT on April 24. We discuss this event because we can examine how a number of possible ejecta signatures are

Table 3. Number of $T_p/T_{ex} \leq 0.5$ Plasma Events Associated with Ejecta Signatures, Cosmic Ray Decreases, and Shocks, and Chance Association Rates

	All Events		Non-HPS		HPS	
	All	≥ 10 hours	All	≥ 10 hours	All	≥ 10 hours
Bidirectional ~ 1 MeV ion flows						
IMP 8/ISEE 3 ^a						
Events	197/554	123/184	161/406	103/153	36/148	20/31
Chance rate	10%	20%	11%	21%	8%	16%
Observed/chance	3.5	3.4	3.5	3.2	3.1	4.1
Helios 1						
Events	152/570	72/169	115/386	52/115	37/184	20/54
Chance rate	9%	23%	16%	24%	9%	21%
Observed/chance	2.9	1.9	3.1	1.9	2.3	1.8
Helios 2						
Events	72/246	39/77	48/159	26/49	24/87	13/28
Chance rate	5%	13%	6%	14%	5%	12%
Observed/chance	5.9	3.9	5.4	3.7	5.9	4.0
ISEE 3 bidirectional solar wind electron heat fluxes						
Events	65/240	44/97	60/188	43/86	5/52	1/11
Chance rate	15%	22%	16%	23%	12%	18%
Observed/chance	1.8	2.1	2.0	2.2	0.84	0.50 $\pm$ 0.50
ISEE 3 electron temperature depression ($T_e \leq 10^5$ K)						
Events	102/226	66/96	75/176	58/85	27/50	8/11
Chance rate	19%	31%	20%	31%	13%	24%
Observed/chance	2.4	2.3	2.1	2.2	4.2	3.1
ISEE 3 local electron temperature depression ^b						
Events	164/232	77/94	124/182	68/84	40/50	9/10
Magnetic cloud						
Events	41/566	24/191	35/401	22/148	6/165	2/43
Chance rate	1.4%	2.0%	1.5%	2.0%	1.3%	1.7%
Observed/chance	5.2	6.3	5.8	7.5	2.8 $\pm$ 0.5	2.7 $\pm$ 1.4
He abundance enhancement ^c						
Events	43/331	30/107	41/215	28/75	2/116	2/32
Chance rate	1.2%	2.8%	1.4%	3.0%	1.0%	2.1%
Observed/chance	10.8	10.0	13.6	12.4	1.8 $\pm$ 0.9	3.0 $\pm$ 1.5
Ion count rate depression >60 MeV/amu ^b						
IMP 8 Events	188/666	109/199	172/464	102/162	16/202	7/37
Helios 1 Events	180/680	83/176	143/436	72/118	37/244	11/58
Helios 2 Events	117/398	56/104	85/232	42/62	32/166	14/42
Neutron monitor decrease $\geq 4\%$						
Events	80/909	47/267	76/660	46/219	4/249	1/48
Chance rate	4.2%	5.5%	4.3%	5.6%	3.9%	5.0%
Observed/chance	2.1	3.2	2.7	3.8	0.4 $\pm$ 0.1	0.4 $\pm$ 0.4
Shock/sc within ~ 1.5 days before event						
Omni						
Events	164/909	96/267	155/659	94/219	9/249	2/48
Chance rate	13%	17%	14%	18%	12%	16%
Observed/chance	1.4	2.1	1.7	2.4	0.3 $\pm$ 0.05	0.3 $\pm$ 0.1
Helios 1						
Events	125/744	58/179	118/475	57/119	7/269	1/60
Chance rate	15%	20%	15%	20%	15%	19%
Observed/chance	1.14	1.7	1.7	2.4	0.18 $\pm$ 0.03	0.09 $\pm$ 0.09
Helios 2						
Events	76/472	34/107	60/270	31/64	16/202	3/43
Chance rate	14%	19%	15%	19%	14%	18%
Observed/chance	1.12	1.7	1.5	2.5	0.56 $\pm$ 0.04	0.39 $\pm$ 0.13

^aBIFs with durations >2 hours. BIFs observed by the GSFC instruments on IMP 8 and ISEE 3 and by the EPAS experiment on ISEE 3 [Marsden *et al.*, 1987] are assumed to be independent observations when calculating the chance association rate.

^bChance association rates not calculated for this signature.

^cOne-hour duration assumed for spot times given by Borini *et al.* [1982].

related to the T_p depression. Figure 9 includes the plasma electron temperature from the LANL instrument on ISEE 3 and the >60 MeV/amu ion count rate from IMP 8. The horizontal bars in the sixth panel indicate a BHF [Gosling *et al.*, 1987] and ~ 1 MeV BIFs inferred using data from the GSFC instruments on ISEE 3 and IMP 8 [Richardson and Reames, 1993]. However, no 35–1000 keV BIF was reported by Marsden

et al. [1987]. A helium abundance enhancement in this ejecta was identified by Zwickl *et al.* [1983]. Note that a magnetic cloud signature is absent.

The vertical dashed lines indicate the extent of the ejecta based primarily on the T_p/T_{ex} depression. The BHFs and BIFs occur in this same region and, taken together, they extend over most of the region. However, none of these signatures alone

Table 4. Chance Association Rates of Ejecta Signatures and Shocks With $T_p/T_{ex} \leq 0.5$ Plasma

Signature	Rate, %	Observed/Chance Ratio
Bidirectional ion flows		
<i>Palmer et al.</i> [1978]	8.5 ± 0.4	7.1 ± 2.4
ISEE 3, 35–1000 keV [<i>Marsden et al.</i> , 1987]	29 ± 2	3.3 ± 0.2
IMP 8 (≥ 4 hours) [<i>Richardson and Reames</i> , 1993]	14 ± 0	3.5 ± 0.1
ISEE 3 (≥ 4 hours) [<i>Richardson and Reames</i> , 1993]	25 ± 0	1.8 ± 0.1
Helios 1 (≥ 4 hours) [<i>Richardson</i> , 1994]	21 ± 1	2.3 ± 0.1
Helios 2 (≥ 4 hours) [<i>Richardson</i> , 1994]	20 ± 2	3.0 ± 0.3
Bidirectional solar wind electron heat fluxes		
<i>Gosling et al.</i> [1987]	23 ± 1	3.2 ± 0.2
<i>Gosling et al.</i> [1990] (large geomagnetic storms)	37 ± 4	2.4 ± 0.3
J. T. Gosling (private communication, 1989)	38 ± 2	1.7 ± 0.1
Plasma electron temperature depressions		
<i>Montgomery et al.</i> [1974]	20 ± 3	3.7 ± 1.4
ISEE 3 (Aug. 1978–Oct. 1982) $T_e \leq 6 \times 10^4$ K	20 ± 1	3.9 ± 0.2
Plasma proton temperature depressions <i>Gosling et al.</i> [1973]	13 ± 3	7.1 ± 1.5
Magnetic clouds		
<i>Klein and Burlaga</i> [1982] and <i>Lepping et al.</i> [1990]	30 ± 1	2.5 ± 0.1
He enhancement		
<i>Hirshberg et al.</i> [1972] and references therein	23 ± 2	3.0 ± 0.4
<i>Borini et al.</i> [1982]	6 ± 0^a	9.2 ± 0.2
<i>Neugebauer and Alexander</i> [1991]	18 ± 2	4.2 ± 0.5
Neutron monitor decreases $\geq 4\%$		
Classes 1 and 3	39 ± 0^b	2.1 ± 0.1
Shocks/sc		
sc (Solar-Geophysical Data Summary Reports, 1965–1991)	21 ± 0	2.4 ± 0
Helios 1 shocks	64 ± 0	0.9 ± 0
Helios 2 shocks	60 ± 0	1.04 ± 0

^aOne-hour duration assumed for spot times given by *Borini et al.* [1982].^bA 1.5-day duration assumed.

identifies the complete ejecta region. The abrupt decrease in the >60 MeV/amu ion count rate at the start of the low- T_p region is typical of entry into ejecta material [*Cane et al.*, 1993, 1994]. A decrease in the ~ 1 MeV particle intensity also occurred at this time [*Cane et al.*, 1994]. At the trailing edge of the low- T_p region, there is a change in the >60 MeV/amu ion recovery rate, as is frequently found on exit from ejecta [*Cane et al.*, 1994]. We suggest therefore that the low- T_p region and particle depression indicate the extent of the ejecta, whereas the bidirectional particle flows occur within this region.

An interesting feature of this ejecta is the difference in the behaviors of the proton and electron temperatures. While both are depressed during the first one third of the proposed ejecta region, T_e recovers and is even enhanced during the remainder of the ejecta. A possible interpretation is that plasma in the trailing edge of the ejecta is better connected to the hot corona than that in the leading edge of the ejecta, where the depression in T_e suggests that thermal conduction from the corona is restricted. One possibility is that the leading edge of the ejecta includes closed field lines disconnected from the Sun, while later, but still inside the ejecta, looped field lines rooted at the Sun are encountered. Since thermal conduction is poorer for protons than for electrons, T_p is depressed in both regions. Another possibility is that T_e increases with time as heat is conducted into the ejecta. We note that the BHF is only reported in the region of depressed T_e , whereas BIFs are observed predominantly, though not exclusively, in the region in which only T_p is depressed. The fact that the >60 MeV/amu ion count rate remains depressed after the end of the BHF, and BIFs continue to be observed, suggests that the ejecta extends beyond the end of the BHF, which therefore does not

indicate the complete ejecta region. Our conclusion from events such as that in Figure 9 is that the interrelationship of ejecta signatures is extremely complex (as was also noted by *Crooker et al.* [1990]) but that regions of abnormally depressed T_p can help to indicate the probable extent of ejecta material. The separation of predominantly electron signatures inside the leading edge of this ejecta and ion signatures inside the trailing edge is the reverse of that in the event discussed by *Crooker et al.* [1990], so that neither pattern of behavior is found universally in ejecta. Before leaving Figure 9 we note the brief T_p depression on April 27 which is virtually coincident with a brief IMP 8 BIF to within the time resolution of the measurements.

The observations in Figure 10, made on February 28 to March 9, 1975, by Helios 1 at 0.39 AU, include another interesting low- T_p region, which passed by on March 4–5. This region was associated with an enhanced magnetic field with a rotation in direction. We suggest that it was an ejecta with a magnetic cloud-like structure. No shock has been reported ahead of this feature (*R. Schwenn*, private communication, 1993), consistent with the low speed of the ejecta compared to the adjacent solar wind. We have little information on other ejecta signatures. The ~ 1 MeV ion fluxes measured by the GSFC Helios 1 instrument were too low to provide information on the presence of BIFs. A striking feature in Figure 10 is the brief, $\sim 9\%$ > 60 MeV/amu ion count rate depression observed by the University of Kiel instrument, which is clearly associated with the region of depressed T_p . We interpret this as a local depression of the cosmic ray density produced by an isolated ejecta which is not associated with a shock. There are two other features to note in Figure 10: The first is the brief T_p depression on March 8, which is apparently associated with an

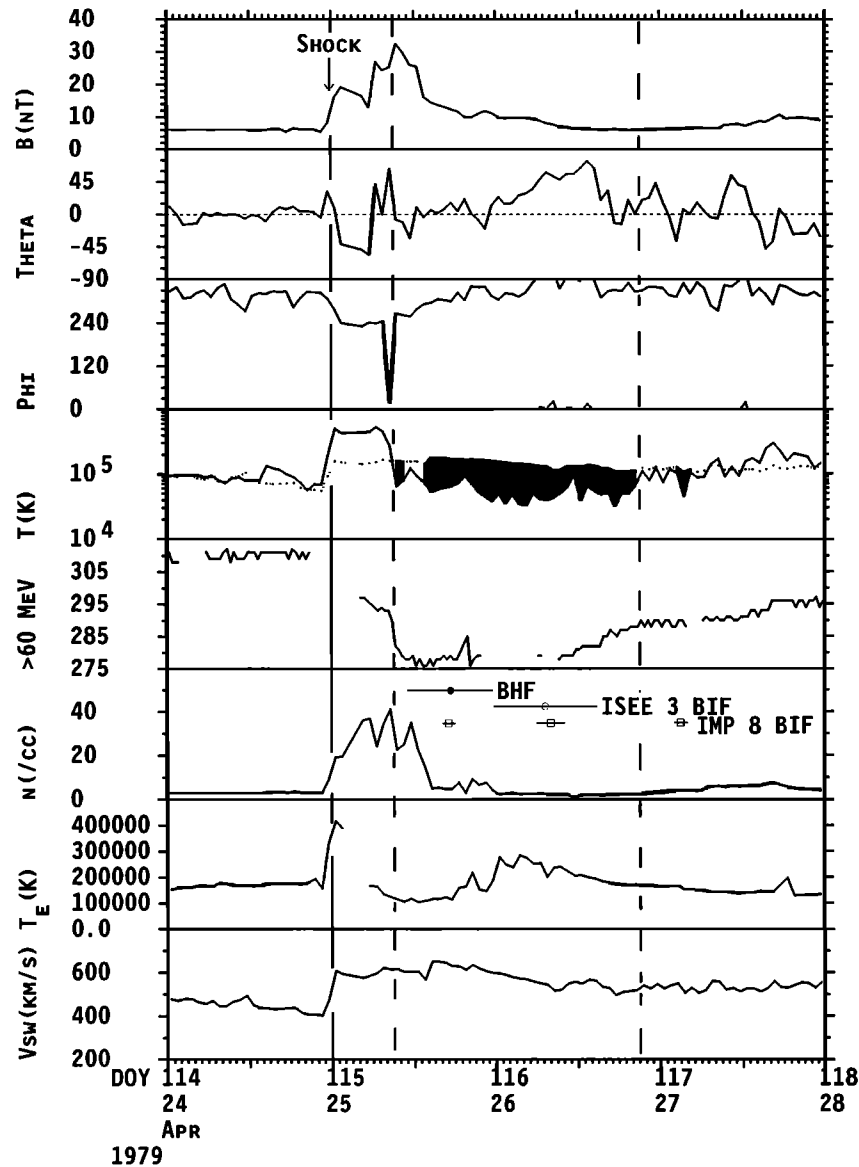


Figure 9. Relationship between the region of depressed T_p and various ejecta signatures following a shock which passed the Earth on April 24, 1979, at 2358 UT. In addition to the parameters in Figure 2, the >60 MeV/amu ion count rate from the anticoincidence guard of the Goddard Space Flight Center (GSFC) instrument on IMP 8 and the electron temperature (T_e) from the Los Alamos National Laboratory experiment on ISEE 3 are shown. The ejecta signatures indicated are bidirectional solar wind electron heat fluxes (BHF) [Gosling *et al.*, 1987] and bidirectional energetic ion flows (BIF) from the GSFC instruments on IMP 8 and ISEE 3.

HPS encounter on the basis of the depressed magnetic field intensity (in this case associated with a magnetic sector boundary crossing, indicating that the heliospheric current sheet is crossed) and enhanced density, slow, solar wind. Note that there is no high-energy particle depression at the HPS, consistent with the conclusions from Table 1 that the HPS is less likely to be associated with such depressions. The second is the four brief T_p depressions, indicated by shading, which are not associated with other obvious signatures in the solar wind data, at least with the available data resolution. The interpretation of these events is unclear, in particular whether they have any association with ejecta or are the result of small-scale local deviations from the T_p - V_{sw} relationship.

Figure 11 shows Omni magnetic field and T_p data together

with ISEE 3 plasma electron temperatures for a 1-month interval (November 6 to December 5, 1979) near solar maximum. This period is shown because observations are nearly complete and there is information available on a wide range of ejecta signatures. In particular, the horizontal bars in the center panel indicate intervals over which BHFs were observed at ISEE 3 [Gosling *et al.*, 1987], and BIFs were detected (at ~ 1 MeV) by the GSFC instruments on ISEE 3 (I3BIF) and IMP 8 (I8BIF), according to the criteria of Richardson and Reames [1993] and (at 35–1000 keV) by the ISEE 3 Energetic Particle Anisotropy Spectrometer (MBIF) [Marsden *et al.*, 1987]. (Note that IMP 8 BIFs are only indicated when the spacecraft was in the solar wind.) Times of sc, which are generally related to the passage of interplanetary shocks, are indicated by vertical lines.

Several intervals are indicated (shaded) where $T_p \leq 0.5 T_{ex}$. Most do not follow shocks, but they are often associated with features, such as magnetic field direction changes, which suggest the presence of discrete solar wind structures. Most of these intervals are associated with BIFs and/or BHF. The electron temperature is also usually depressed in the regions of low T_p and BIFs or BHFs, though this is not always the case.

Our impression from this and other such periods is that no one signature provides a means of identifying, unambiguously and reliably, all ejecta regions in the solar wind. In particular, neither the reported BIFs nor BHFs alone identify all of the regions in which other ejecta signatures are observed. For example, there is no reported BHF in the low (proton and electron) temperature plasma event of November 25–26 in which BIFs are observed by two instruments on ISEE 3 and by IMP 8. On the other hand, several of the GSFC IMP 8 and ISEE 3 BIFs are not associated with clear T_p depressions. The Marsden *et al.* [1987] BIF events appear to be a subset of these events, as was also concluded for BIFs following strong shocks by Richardson and Cane [1993]. The observations in Figure 11 suggest that T_p may provide a more comprehensive indication of the presence of ejecta than the reported bidirectional particle flows. The observations indicate the point made by Richardson and Cane [1993] that ideally as many signatures as possible should be examined when trying to identify ejecta in the solar wind.

6. Summary and Discussion

In this study we have considered further the suggestion of Gosling *et al.* [1973] and the recent studies of Richardson and Cane [1993] and Cane *et al.* [1994] that plasma in which T_p is significantly less than that expected for normal solar wind expansion may indicate the presence of ejecta material in the solar wind. We have identified ~2000 low- T_p intervals in the NSSDC Omni and Helios plasma databases between 1965 and 1991. Around 30% of these intervals are encounters with the “quiet” heliospheric plasma sheet. We will summarize the nature of these events below. The non-HPS intervals have the following characteristics, several of which suggest a reasonable association with ejecta:

1. Non-HPS low- T_p events have durations of ~1 to ~80 hours, suggesting scale sizes ~0.01 to 0.8 AU if convected past the observer at ~500 km/s. The large-scale sizes attained by some events are consistent with the expanding CME structures extending over many tens of degrees in latitude observed near the Sun by coronagraphs [e.g., Hundhausen, 1993].
2. The mean event duration increases with heliocentric distance and is consistent with an expansion rate of the order of the Alfvén speed. We note that Klein and Burlaga [1982] suggested that ejecta (in their case specifically magnetic clouds) may expand at speeds less than or the order of the Alfvén speed.
3. Around 25% of non-HPS events closely follow interplanetary shocks or geomagnetic storm sudden commencements. This association rate is similar to those found for BHFs [Gosling *et al.*, 1987] and for BIFs [Richardson and Reames, 1993; Richardson, 1994]. However, the association rate is not significantly greater than would be expected from chance except for events with durations of 10 hours or more, for which the association rate increases to ~45%. The reasonable implication is that it is the longer-duration (and hence larger-scale) events which generate shocks.

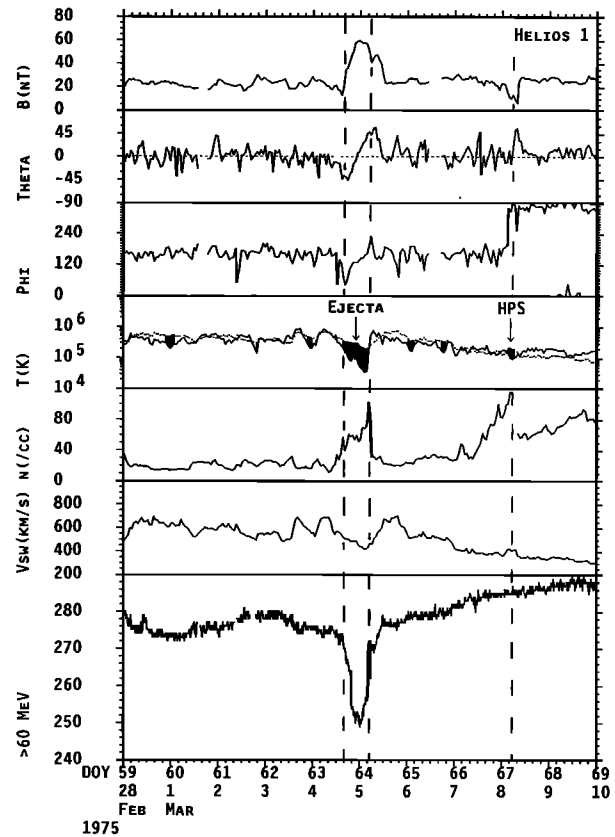


Figure 10. Helios 1 solar wind and energetic particle data for a 10-day period in February–March, 1975. We interpret the region of depressed T_p accompanied by a magnetic field enhancement and rotation as an ejecta with a magnetic cloud-like structure. There is no shock preceding this ejecta, which is associated with a local ~9% decrease in the >60 MeV/amu ion count rate.

4. The number of events observed falls off with the event duration, as is found for other ejecta signatures [see Richardson and Reames, 1993, Figure 11]. However, this contrasts with expectations based on the latitudinal widths of CMEs [Hundhausen, 1993], which show a decrease in frequency at small angular widths. On the other hand, an analysis by Jackson [1992] suggests a high rate for small CMEs.

5. The yearly averaged occurrence rate of plasma with $T_p/T_{ex} \leq 0.5$ is correlated with solar activity levels, with low- T_p plasma being observed more frequently around solar maximum. This correlation is particularly clear for the occurrence rate of low- T_p plasma in non-HPS events, which also shows a good correlation with the CME rate at the Sun observed by coronagraphs, where such observations are available. The T_p data may provide information on the CME rate during the period before the late 1970s, when there were few coronagraph observations.

6. The estimated event rate is consistent with the expected ejecta rate at the Earth inferred by Gosling *et al.* [1992] from SMM CME observations, in particular if the 70% SMM detection efficiency is taken into account. The similar event rates at the Helios spacecraft and at the Earth suggest that low- T_p regions continue to move out away from the Sun.

7. Low-temperature events show significant associations (greater than chance) with other possible signatures of ejecta.

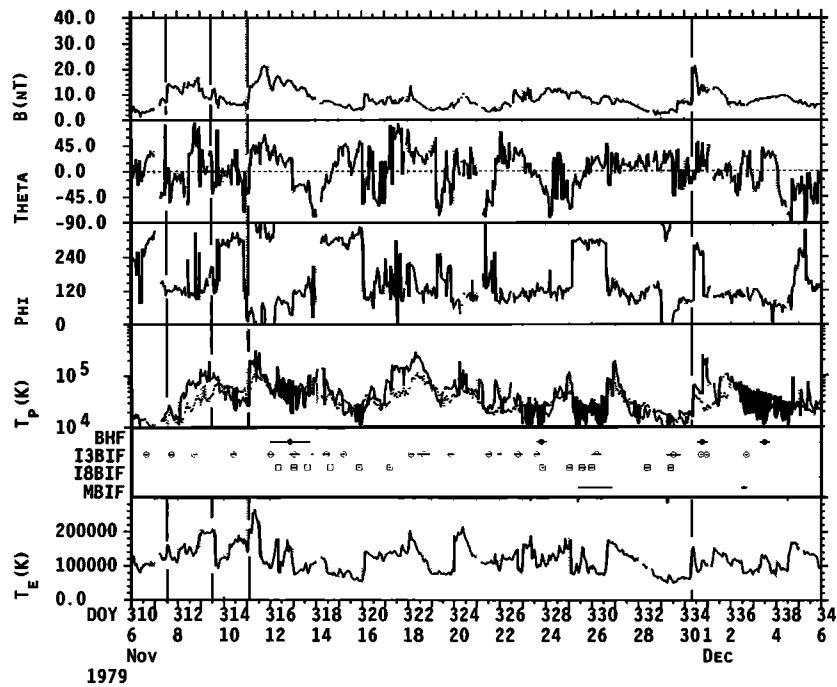


Figure 11. Solar wind data for November 6 to December 5, 1979. The fifth panel shows intervals in which bidirectional particle flows are observed, including BHF [Gosling *et al.*, 1987], ~ 1 MeV BIFs observed by the GSFC instruments on IMP 8 and ISEE 3 (I8BIF, I3BIF), and 35–1000 keV BIFs observed by the energetic particle anisotropy spectrometer instrument on ISEE 3 (MBIF) [Marsden *et al.*, 1987]. Solid vertical lines indicate sc.

In particular, typically half or more of the events listed in various studies of other signatures are associated with plasma with $T_p/T_{ex} \leq 0.5$.

8. As has been noted before [e.g., Zwickl *et al.*, 1983; Crooker *et al.*, 1990; Richardson and Reames, 1993], ejecta signatures in an ejecta usually do not coincide exactly and may not be present throughout the whole ejecta. The examples discussed above suggest that these signatures tend to occur within the region of low- T_p plasma. This, together with the close association between energetic particle depressions and low- T_p plasma we have noted above and elsewhere, suggests that low- T_p plasma may be a better indication of the duration of the ejecta.

Around 30% of the low- T_p events identified are associated with encounters with the quiet heliospheric plasma sheet. The following characteristics have been observed:

1. They are found at all phases of the solar cycle but are most evident following solar minimum during the increasing phase of solar activity when the heliospheric current sheet tilt angle is small [e.g., Cliver, 1993], so that the HPS is more likely to be encountered in the ecliptic.
2. They have durations of ~ 1 to 50 hours with a mean duration of ~ 7 hours, which for corotating structures suggests longitudinal widths of $\sim 4^\circ$.
3. These events rarely follow strong shocks. This lack of association may in part be a selection effect since, following a strong shock, the solar wind speed is likely to be enhanced so that the low-speed solar wind criterion for identifying HPS events will not be met.
4. The association with possible ejecta signatures is complex. They show higher than chance associations with bidirectional ~ 1 MeV ions and electron temperature depressions.

However, they tend not to be observed in association with bidirectional solar wind electron heat fluxes, high-energy particle decreases, helium abundance enhancements, and magnetic cloud signatures. While the observation of some ejecta signatures in the HPS provides support for the proposal of Crooker *et al.* [1993] that ejecta may occur as an intrinsic part of the HPS, the absence of other signatures which are believed to indicate the presence of ejecta suggests that ejecta may not be present or that conditions inside ejecta moving out through the quiet HPS differ from those in fast ejecta.

Thus in conclusion we suggest that the identification of regions with abnormally low T_p is a useful method of identifying (though not uniquely) possible ejecta material in the solar wind. In particular, because near-Earth observations of T_p already exist for 2 1/2 solar cycles and similar data are available from spacecraft elsewhere in the heliosphere, the data have the potential of providing information on the occurrence of ejecta both over a long time base and simultaneously from multiple spacecraft in order to investigate ejecta structure. We suggest that low- T_p plasma appears to be a more general signature of ejecta than bidirectional solar wind electron heat fluxes or bidirectional ion flows, while stressing that ideally, as many signatures as possible should be examined when trying to identify ejecta.

Appendix: Calculation of Chance Association Rates

The chance association rates for the groups of events in Table 1 are calculated as follows: The probability of one low- T_p region of duration t_T overlapping by chance with a single period of duration t_s in which a particular ejecta signa-

ture is observed is $P_s = (t_s + t_T)/t_o$, where t_o is the total duration of the observation interval (typically several years). If there are additional signature periods in this observation interval (typically there will be many tens of such periods), then the probability of a chance association between the low- T_p region and any signature period is $P_s = \Sigma(t_s + t_T)/t_o$, summing over all signature periods. In this observation period, there will be many (tens or hundreds of) low- T_p regions with a range of durations. The appropriate parameter to compare with the association rates in Table 1 is the probability of a chance association (as above), averaged over all the low- T_p events in this observation period, taking into account their different durations t_T . These rates are shown in Table 3 together with the number of events included in each group of events, the number of these events associated with the ejecta signature, and the ratio of the observed and chance association rates. The errors in the chance association rates are generally $<1\%$ and are not shown in the table. For some signatures, no comprehensive list of events is currently available, so the chance association rates cannot be estimated. Note that comparison of the observed and chance association rates will still indicate the significance of the observed association rate even if data gaps are present in the signature observations. For example, if the data coverage were 50%, then only around half of the signature events present in the observation period are likely to have been identified. However, the probability of a chance association, which is calculated from the number of signature events identified, will be reduced by the same proportion.

Table 4 shows the chance association rates relating to Table 2. To calculate the chance association rate, the probability of one signature interval with duration t_s being associated with any low- T_p region is given by $P_T = (1/at_o) \Sigma(t_s + t_T)$, where the summation is over all low- T_p events with durations t_T detected in the observation period t_o , and $a(\leq 1)$ is the T_p data coverage. To obtain the chance association rate, the mean probability of an association is calculated for all the signature events, taking into account their durations. Note that in this case the T_p data coverage ($a \leq 1$) must be taken into account, since if coverage is incomplete, the number of low- T_p events actually present in the observation period, and hence the probability of an association, will be underestimated.

Acknowledgments. We acknowledge the use of data from the tape of Helios merged hourly averaged field (P.I. F.M. Neubauer) and plasma (P.I. H. Rosenbauer) data (NSSDC ID 74-097A-09B) from the Omni on-line database and from the tape of plasma electron data from the Los Alamos National Laboratory instrument on ISEE 3 (P.I. S.J. Bame; NSSDC ID 78-079A-01K), all provided by the National Space Science Data Center. We thank G. Wibberenz for providing the University of Kiel Helios particle data and R. Schwenn for supplying an unpublished Helios shock list. Reference to an unpublished list of bidirectional solar wind electron heat flux events compiled by J. T. Gosling is gratefully acknowledged. HVC was supported at GSFC by a contract with Universities Space Research Association. The work of IGR was supported by NASA grant NGR 21-002-316.

The Editor thanks V. J. Pizzo, J. Kóta, and another referee for their assistance in evaluating this paper.

References

- Borriani, G., J. T. Gosling, S. J. Bame, W. C. Feldman, and J. M. Wilcox, Solar wind helium and hydrogen structure near the heliospheric current sheet: A signal of coronal streamers at 1 AU, *J. Geophys. Res.*, **86**, 4565, 1981.
- Borriani, G., J. T. Gosling, S. J. Bame, and W. C. Feldman, Helium

- abundance enhancements in the solar wind, *J. Geophys. Res.*, **87**, 7370, 1982.
- Bruno, R., U. Villante, and A. Stecca, Selected solar wind parameters at 1 AU through two solar activity cycles, *Ann. Geophys.*, **12**, 105, 1994.
- Burlaga, L. F., and K. W. Ogilvie, Solar wind temperature and speed, *J. Geophys. Res.*, **78**, 2028, 1973.
- Burlaga, L. F., D. H. Fairfield, M. D. Montgomery, and S. J. Bame, Energy transfer at colliding streams in the solar wind, *Astrophys. J.*, **164**, 137, 1971.
- Cane, H. V., The large-scale structure of flare-associated interplanetary shocks, *J. Geophys. Res.*, **93**, 1, 1988.
- Cane, H. V., and I. G. Richardson, Cosmic ray decreases and solar wind disturbances during late October 1989, *J. Geophys. Res.*, **100**, 1755, 1995.
- Cane, H. V., I. G. Richardson, and T. T. von Rosenvinge, Cosmic ray decreases and particle acceleration in 1978–1982 and the associated solar wind structures, *J. Geophys. Res.*, **98**, 13,295, 1993.
- Cane, H. V., I. G. Richardson, T. T. von Rosenvinge, and G. Wibberenz, Cosmic ray decreases and shock structure: A multispacecraft study, *J. Geophys. Res.*, **99**, 21,429, 1994.
- Cliver, E. W., The shapes of galactic cosmic ray intensity maxima and the evolution of the heliospheric current sheet, *J. Geophys. Res.*, **98**, 17,435, 1993.
- Couzens, D. A., and J. H. King, Interplanetary medium data book, *Rep. NSSDC/WDC-A-R&S 86-04*, suppl. 3, 1977–1985, Natl. Space Sci. Data Cent., Greenbelt, Md., 1986.
- Crooker, N. U., J. T. Gosling, E. J. Smith, and C. T. Russell, A bubblelike coronal mass ejection flux rope in the solar wind, in *Physics of Magnetic Flux Ropes*, *Geophys. Monogr. Ser.*, vol. 58, edited by C. T. Russell, E. R. Priest, and L. C. Lee, p. 365, AGU, Washington, D. C., 1990.
- Crooker, N. U., G. L. Siscoe, S. Shodhan, D. F. Webb, J. T. Gosling, and E. J. Smith, Multiple heliospheric current sheets and coronal streamer belt dynamics, *J. Geophys. Res.*, **98**, 9371, 1993.
- Didato, L., G. Moreno, C. Signorini, and K. W. Ogilvie, Long-term variations of the solar wind protons parameters, *J. Geophys. Res.*, **79**, 5095, 1974.
- Farrugia, C. J., L. F. Burlaga, V. A. Osherovich, I. G. Richardson, M. P. Freeman, R. P. Lepping, and A. J. Lazarus, A study of an expanding interplanetary magnetic cloud and its interaction with the Earth's magnetosphere: The interplanetary aspect, *J. Geophys. Res.*, **98**, 7621, 1993a.
- Farrugia, C. J., I. G. Richardson, L. F. Burlaga, V. A. Osherovich, and R. P. Lepping, Simultaneous observations of solar MeV particles in a magnetic cloud and in the Earth's northern tail lobe: Implications for the global field line topology of magnetic clouds and for the entry of solar particles into the magnetosphere during cloud passage, *J. Geophys. Res.*, **98**, 15,497, 1993b.
- Feldman, W. C., J. R. Asbridge, S. J. Bame, E. E. Fenimore, and J. T. Gosling, The solar origins of solar wind interstream flows: Near equatorial coronal streamers, *J. Geophys. Res.*, **86**, 5408, 1981.
- Feynman, J., Solar cycle and long-term changes in the solar wind, *Rev. Geophys.*, **21**, 338, 1983.
- Freeman, J. W., and R. E. Lopez, The cold solar wind, *J. Geophys. Res.*, **90**, 9885, 1985.
- Gosling, J. T., Coronal mass ejections and magnetic flux ropes in interplanetary space, in *Physics of Magnetic Flux Ropes*, *Geophys. Monogr. Ser.*, vol. 58, edited by C. T. Russell, E. R. Priest, and L. C. Lee, p. 343, AGU, Washington, D. C., 1990.
- Gosling, J. T., V. Pizzo, and S. J. Bame, Anomalous low proton temperatures in the solar wind following interplanetary shock waves: Evidence for magnetic bottles?, *J. Geophys. Res.*, **78**, 2001, 1973.
- Gosling, J. T., G. Borriani, J. R. Asbridge, S. J. Bame, W. C. Feldman, and R. T. Hansen, Coronal streamers in the solar wind at 1 AU, *J. Geophys. Res.*, **86**, 5438, 1981.
- Gosling, J. T., D. N. Baker, S. J. Bame, W. C. Feldman, R. D. Zwickl, and E. J. Smith, Bidirectional solar wind electron heat flux events, *J. Geophys. Res.*, **92**, 8519, 1987.
- Gosling, J. T., S. J. Bame, D. J. McComas, and J. L. Phillips, Coronal mass ejections and large geomagnetic storms, *Geophys. Res. Lett.*, **17**, 901, 1990.
- Gosling, J. T., D. J. McComas, J. L. Phillips, and S. J. Bame, Counterstreaming solar wind halo electron events: Solar cycle variations, *J. Geophys. Res.*, **97**, 6531, 1992.
- Gosling, J. T., S. J. Bame, D. J. McComas, J. L. Phillips, B. E. Gold-

- stein, and M. Neugebauer, The speeds of coronal mass ejections in the solar wind at mid heliographic latitude: Ulysses, *Geophys. Res. Lett.*, **21**, 1109, 1994.
- Hirshberg, J., S. J. Bame, and D. E. Robbins, Solar flares and solar wind helium enrichments: July 1965–July 1967, *Sol. Phys.*, **23**, 467, 1972.
- Hundhausen, A. J., Interplanetary shock waves and the structure of solar wind disturbances, in *Solar Wind*, edited by C. P. Sonett, P. J. Coleman Jr., and J. M. Wilcox, *NASA Spec. Publ.*, **SP-308**, 393, 1972.
- Hundhausen, A. J., The sizes and locations of coronal mass ejections: SMM observations from 1980 and 1984–1989, *J. Geophys. Res.*, **98**, 13,177, 1993.
- Jackson, B. V., Remote sensing observations of mass ejections and shocks in interplanetary space, in *Eruptive Solar Flares*, edited by Z. Svestka, B. Jackson, and M. Machado, p. 248, Springer-Verlag, New York, 1992.
- Kahler, S. W., and D. V. Reames, Probing the magnetic topologies of magnetic clouds by means of solar energetic particles, *J. Geophys. Res.*, **96**, 9419, 1991.
- Klein, L. W., and L. F. Burlaga, Interplanetary magnetic clouds at 1 AU, *J. Geophys. Res.*, **87**, 613, 1982.
- Lepping, R. P., J. A. Jones, and L. F. Burlaga, Magnetic field structure of interplanetary magnetic clouds at 1 AU, *J. Geophys. Res.*, **95**, 11,957, 1990.
- Lockwood, J. A., List of Forbush decreases 1954–1990, *Solar-Geophysical Data*, vol. 549, Part 1, p. 154, 1990.
- Lopez, R. E., Solar cycle invariance in solar wind proton temperature relationships, *J. Geophys. Res.*, **92**, 11,189, 1987.
- Lopez, R. E., and J. W. Freeman, The solar wind proton temperature-velocity relationship, *J. Geophys. Res.*, **91**, 1701, 1986.
- Marsden, R. G., T. R. Sanderson, C. Tranquille, K.-P. Wenzel, and E. J. Smith, ISEE 3 observations of low-energy proton bidirectional events and their relation to isolated interplanetary structures, *J. Geophys. Res.*, **92**, 11,009, 1987.
- Montgomery, M. D., J. R. Asbridge, S. J. Bame, and W. C. Feldman, Solar wind electron temperature depressions following some interplanetary shock waves: Evidence for magnetic merging?, *J. Geophys. Res.*, **79**, 3103, 1974.
- Neugebauer, M., and C. J. Alexander, Shuffling footpoints and magnetohydrodynamic discontinuities in the solar wind, *J. Geophys. Res.*, **96**, 9409, 1991.
- Palmer, I. D., F. R. Allum, and S. Singer, Bidirectional anisotropies in solar cosmic ray events: Evidence for magnetic bottles, *J. Geophys. Res.*, **83**, 75, 1978.
- Phillips, J. L., J. T. Gosling, D. J. McComas, S. J. Bame, and W. C. Feldman, Quantitative analysis of bidirectional electron heat fluxes within coronal mass ejections at 1 AU, in *Solar Wind Seven*, edited by E. Marsch and R. Schwenn, p. 651, Pergamon, Tarrytown, N. Y., 1992.
- Richardson, I. G., A survey of bidirectional ≥ 1 MeV ion flows during the Helios 1 and Helios 2 missions: Observations from the Goddard Space Flight Center instruments, *Astrophys. J.*, **420**, 926, 1994.
- Richardson, I. G., and H. V. Cane, Signatures of shock drivers in the solar wind and their dependence on the solar source location, *J. Geophys. Res.*, **98**, 15,295, 1993.
- Richardson, I. G., and D. V. Reames, Bidirectional ~ 1 MeV/amu ion intervals in 1973–1991 observed by the Goddard Space Flight Center instruments on IMP 8 and ISEE 3/ICE, *Astrophys. J. Suppl. Ser.*, **85**, 411, 1993.
- Totten, T. L., J. W. Freeman, and S. Arya, An empirical determination of the polytropic index for the free-streaming solar wind using Helios 1 data, *J. Geophys. Res.*, **100**, 13, 1995.
- Webb, D. F., and R. A. Howard, The solar cycle variation of the occurrence rate of coronal mass ejections and the solar wind mass flux, *J. Geophys. Res.*, **99**, 4201, 1994.
- Winterhalter, D., E. J. Smith, M. E. Burton, N. Murphy, and D. J. McComas, The heliospheric plasma sheet, *J. Geophys. Res.*, **99**, 6667, 1994.
- Zwickl, R. D., J. R. Asbridge, S. J. Bame, W. C. Feldman, J. T. Gosling, and E. J. Smith, Plasma properties of driver gas following interplanetary shocks observed by ISEE 3, Solar Wind Five, *NASA Conf. Publ.*, **CP-2280**, 711, 1983.

H. V. Cane and I. G. Richardson, Laboratory for High Energy Astrophysics, Code 661, NASA Goddard Space Flight Center, Greenbelt, MD 20771. (e-mail: richardson@lhea.gsfc.nasa.gov;cane@nssdca.gsfc.nasa.gov)

(Received October 27, 1994; revised August 24, 1995; accepted August 29, 1995.)